深蓝学院
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DSO
论文带读



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研究方向：机器人与认知系统

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SLAM，SFM

Github：https://github.com/jlyw1017

知乎：https://zhuanlan.zhihu.com/c_147898137



DSO概述

光度校核

优化模型

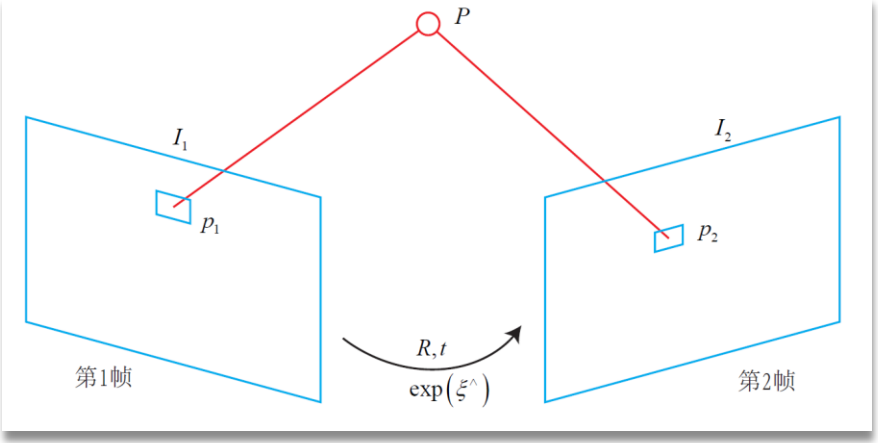
窗口优化

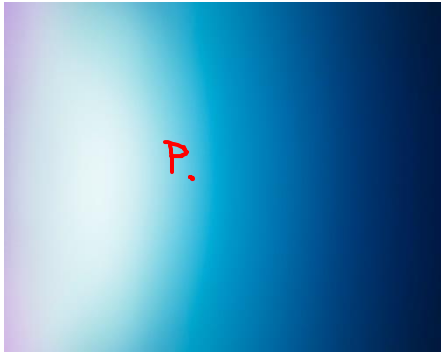
帧管理

点管理

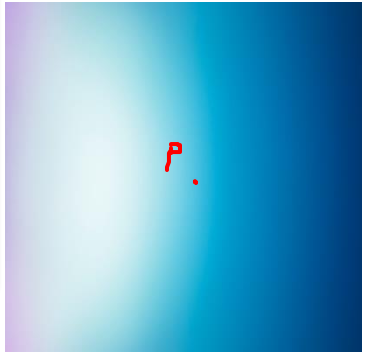
背景知识

直接法

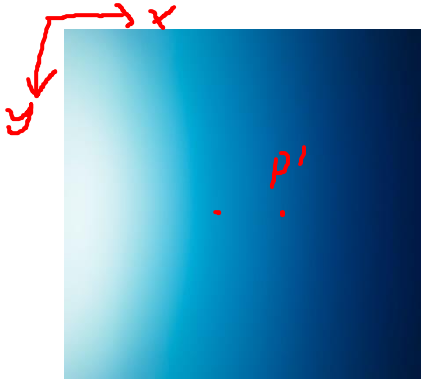








相机1



相机2

背景知识

最小二乘优化问题

$$f(x) = \frac{1}{2}\sum_{j=1}^{m} r_j^2(x),$$

$$J(x) = \left[\frac{\partial r_j}{\partial x_i}\right]_{\substack{j=1,2,\ldots,m \\ i=1,2,\ldots,n}} = \begin{bmatrix} \nabla r_1(x)^T \\ \nabla r_2(x)^T \\ \vdots \\ \nabla r_m(x)^T \end{bmatrix},$$

$$\nabla f(x) = \sum_{j=1}^{m} r_j(x)\nabla r_j(x) = J(x)^T r(x), \quad \text{一阶导}$$

$$\frac{\partial}{\partial \Delta \mathbf{x}}\left(F(\mathbf{x}) + \mathbf{J}\Delta\mathbf{x} + \frac{1}{2}\Delta\mathbf{x}^\top \mathbf{H}\Delta\mathbf{x}\right) = \mathbf{J}^\top + \mathbf{H}\Delta\mathbf{x} = \mathbf{0}$$

$$\nabla^2 f(x) = \sum_{j=1}^{m} \nabla r_j(x)\nabla r_j(x)^T + \sum_{j=1}^{m} r_j(x)\nabla^2 r_j(x) \quad \text{二阶导}$$

$$= J(x)^T J(x) + \sum_{j=1}^{m} r_j(x)\nabla^2 r_j(x).$$

第二项通常比较小

DSO综述

1、直接稀疏法，基于优化

2、两种优化参数

3、达到实时性手段

4、光度标定模型

Abstract

程计We propose a novel direct sparse visual odometry formulation. It combines a fully direct probabilistic model (minimizing a photometric error) with consistent, joint optimization of all model parameters, including geometry – represented as inverse depth in a reference frame – and camera motion. This is achieved in real time by omitting the smoothness prior used in other direct methods and instead sampling pixels evenly throughout the images. Since our method does not depend on keypoint detectors or descriptors, it can naturally sample pixels from across all image regions that have intensity gradient, including edges or smooth intensity variations on mostly white walls. The proposed model integrates a full photometric calibration, accounting for exposure time, lens vignetting, and non-linear response functions. We thoroughly evaluate our method on three different datasets comprising several hours of video. The experiments show that the presented approach significantly outperforms state-of-the-art direct and indirect methods in a variety of real-world settings, both in terms of tracking accuracy and robustness.

DSO综述

1、最大后验概率

2、非直接法简介

arXiv:16

背景，
SLAM应用广
blablabla

1. Introduction

Simultaneous localization and mapping (SLAM) and visual odometry (VO) are fundamental building blocks for many emerging technologies – from autonomous cars and UAVs to virtual and augmented reality. Realtime methods for SLAM and VO have made significant progress in recent years. While for a long time the field was dominated by feature-based (indirect) methods, in recent years a number of different approaches have gained in popularity, namely *direct* and *dense* formulations.

Direct vs. Indirect. Underlying all formulations is a probabilistic model that takes noisy measurements $\mathbf{Y}$ as input and computes an estimator $\mathbf{X}$ for the unknown, hidden model parameters (3D world model and camera

*This work was supported by the ERC Consolidator Grant "3D Reloaded" and by a Google Faculty Research Award.

mulated over the course of the trajectory. The bottom row shows some video frames.

最大后验概率

motion). Typically a Maximum Likelihood approach is used, which finds the model parameters that maximize the probability of obtaining the actual measurements, i.e., $\mathbf{X}^* := \text{argmax}_{\mathbf{X}} P(\mathbf{Y}|\mathbf{X})$.

Indirect法

Indirect methods then proceed in two steps. First, the raw sensor measurements are pre-processed to generate an intermediate representation, solving part of the overall problem, such as establishing correspondences. Second, the computed intermediate values are interpreted as noisy measurements $\mathbf{Y}$ in a probabilistic model to estimate geometry and camera motion. Note that the first step is typically approached by extracting and matching a sparse set of keypoints – however other options exist, like establishing correspondences in the form of dense, regularized optical flow. It can also include methods that extract and match parametric

分两步

1.找几何对应关系

2.根据几何对应关系求解运动

比如特征点法

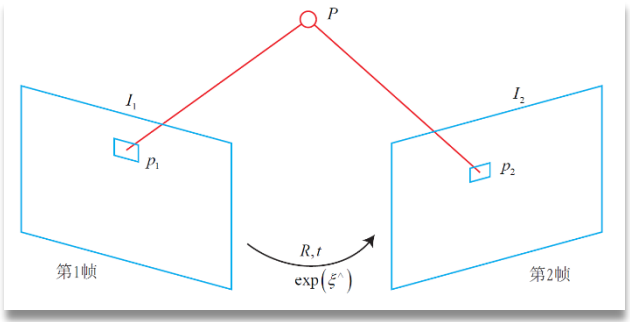
比如其他一些取点线的方法

https://zhuanlan.zhihu.com/p/31870182

DSO综述

直接法简介

小讨论下直接，非直接；几何非几何



representations of other geometric primitives, such as line- or curve-segments.

直接法

不提取中间量

Direct methods skip the pre-processing step and directly use the actual sensor values – light received from a certain direction over a certain time period – as measurements $\mathbf{Y}$ in a probabilistic model.

In the case of passive vision, the direct approach thus optimizes a *photometric error*, since the sensor provides photometric measurements. Indirect methods on the other hand optimize a *geometric error*, since the pre-computed values – point-positions or flow-vecors – are geometric quantities. Note that for other sensor modalities like depth cameras or laser scanners (which directly measure geometric quantities) direct formulations may also optimize a geometric error.

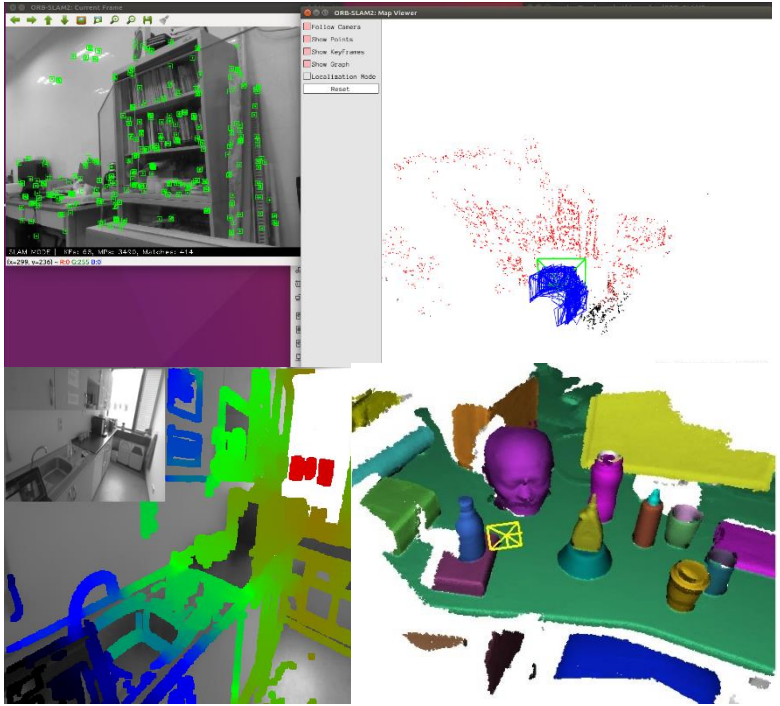
DSO综述

- Sensors are also classified as:
  - **Active sensors:** emit some form of energy into the environment and then measures the return to understand the environment (e.g., infrared sensor, laser range finders.).
    - More robust, less efficient
  - **Passive sensors:** monitor the environment without affecting it, or in other words receive energy already in the environment (e.g., gyroscope, vision camera)
    - Less intrusive, but depends on environment, e.g. temperature probes, cameras.

KIT， ROBOTIKIII H2T 版权所有

DSO综述

稠密 VS 非稠密



三两者及

半稠密的简介

一个重点！

两者的主要区别

是几何先验

Dense vs. Sparse. Sparse methods use and reconstruct only a selected set of independent points (traditionally corners), whereas dense methods attempt to use and reconstruct all pixels in the 2D image domain. Intermediate approaches (semi-dense) refrain from reconstructing the complete surface, but still aim at using and reconstructing a (largely connected and well-constrained) subset.

Apart from the extent of the used image region however, a more fundamental – and consequential – difference lies in the addition of a geometry prior. In the sparse formulation, there is no notion of neighborhood, and geometry parameters (keypoint positions) are conditionally independent given the camera poses & intrinsics[1]. Dense (or semi-dense) approaches on the other hand exploit the connectedness of the used image region to formulate a geometry prior, typically favouring smoothness. In fact, such a prior is necessarily required to make a dense world model observable from passive vision alone. In general, this prior is formulated directly in the form of an additional log-likelihood energy term [26, 21, 22].

DSO综述 Motivation 详细讲为什么直接法，为什么稀疏

$$2x + 2y = 6$$
$$x + 2y = 5$$

Ax = b

JᵀJ E

直接 + 稀疏

直接着重强调了下曝光和伽马校正

稀疏着重强调了下没有几何先验

几何先验是啥？哪来的？为什么几何先验不好？

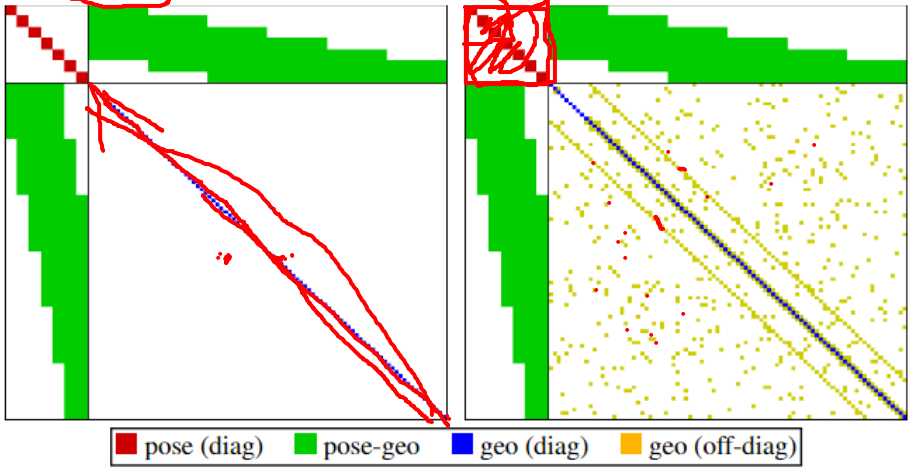


Figure 2. **Sparse vs. dense Hessian structure.** Left: Hessian structure of sparse bundle adjustment: since the geometry-geometry block is diagonal, it can be solved efficiently using the Schur complement. Right: A geometry prior adds (partially unstructured) geometry-geometry correlations – the resulting system is hence not only much larger, but also becomes much harder to solve. For simplicity, we do not show the global camera intrinsic parameters.

Direct Sparse Model

Notation

- Scalars t
- Vectors x
- Matrix H
- Function I
- Poses SE3 Ti se3 xi

$$\boldsymbol{x}_i \boxplus \mathbf{T}_i := e^{\widehat{\boldsymbol{x}_i}} \cdot \mathbf{T}_i. \qquad (1)$$

Direct Sparse Model

2.1.2、Photometric Camera Calibration

- $$I_i(\mathbf{x}) = G\big(t_i V(\mathbf{x}) B_i(\mathbf{x})\big), \qquad (2)$$
- $$I_i'(\mathbf{x}) := t_i B_i(\mathbf{x}) = \frac{G^{-1}(I_i(\mathbf{x}))}{V(\mathbf{x})}. \qquad (3)$$



V 修正晕影 vignetting

Bi irradiance
Ii intensity
ti exposure time

https://www.zhihu.com/question/53080536

J. Engel, V. Usenko, and D. Cremers. A photometrically calibrated benchmark for monocular visual odometry. InarXiv preprint arXiv, 2016

优化模型

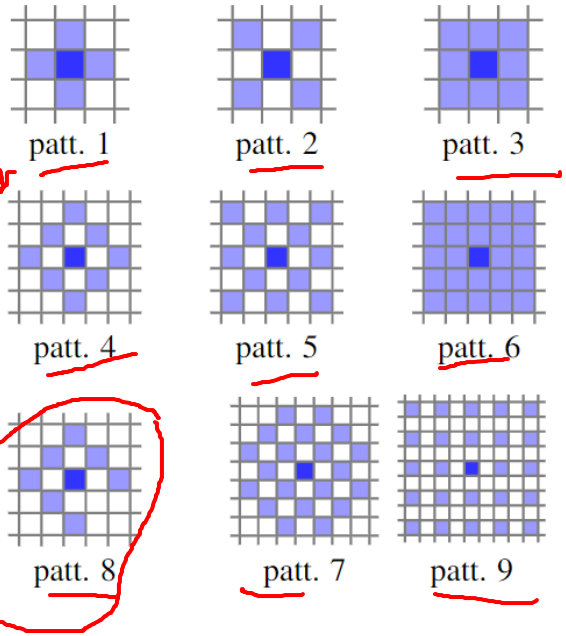
2.2. Model Formulation

$$E_{\mathbf{p}j} := \sum_{\mathbf{p}\in\mathcal{N}_\mathbf{p}} w_\mathbf{p} \left\| \left(I_j[\mathbf{p}'] - b_j\right) - \frac{t_j e^{a_j}}{t_i e^{a_i}} \left(I_i[\mathbf{p}] - b_i\right) \right\|_\gamma , \quad (4)$$

Np 一个邻域 pattern

其中 wp $$w_\mathbf{p} := \frac{c^2}{c^2 + \|\nabla I_i(\mathbf{p})\|_2^2}, \quad (7)$$

B，t，exp(a) 光度修正



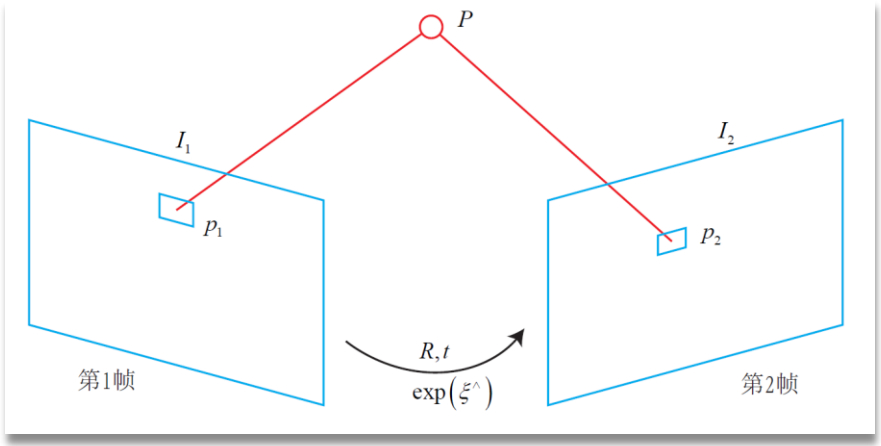
优化模型

2.2. Model Formulation

$$\mathbf{p}' = \Pi_{\mathbf{c}}\left(\mathbf{R}\,\Pi_{\mathbf{c}}^{-1}(\mathbf{p}, d_{\mathbf{p}}) + \mathbf{t}\right), \tag{5}$$

Πc为相机投影，c为内参

$$\begin{bmatrix}\mathbf{R} & \mathbf{t} \\ 0 & 1\end{bmatrix} := \mathbf{T}_j \mathbf{T}_i^{-1}. \tag{6}$$



思考：为什么用逆深度？Point Dimensionality



J. Civera, A. Davison, and J. Montiel. Inverse depth parametrization for monocular SLAM. Transactions on Robotics, 24(5):932–945, 2008

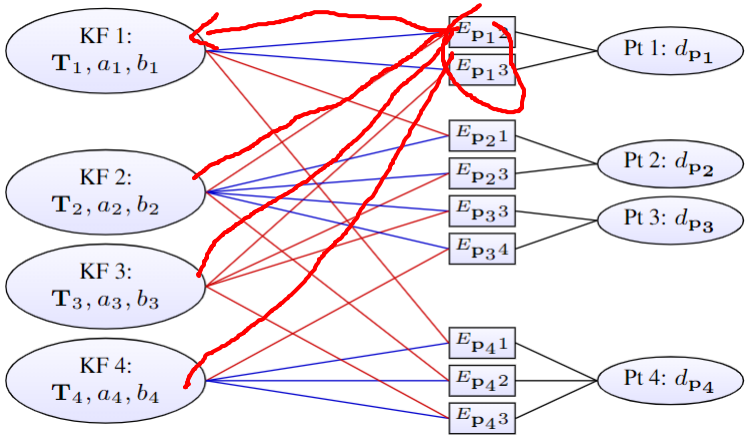
优化模型

2.2. Model Formulation

蓝线：所谓主导帧，就是前面说的i帧

红线：点被投影过去的帧

思考，某个点，是否可以即是本帧所主导的一个点，又是其他点投影的目标位置呢?
论文中关于Consistency的讨论





边缘化舒尔补

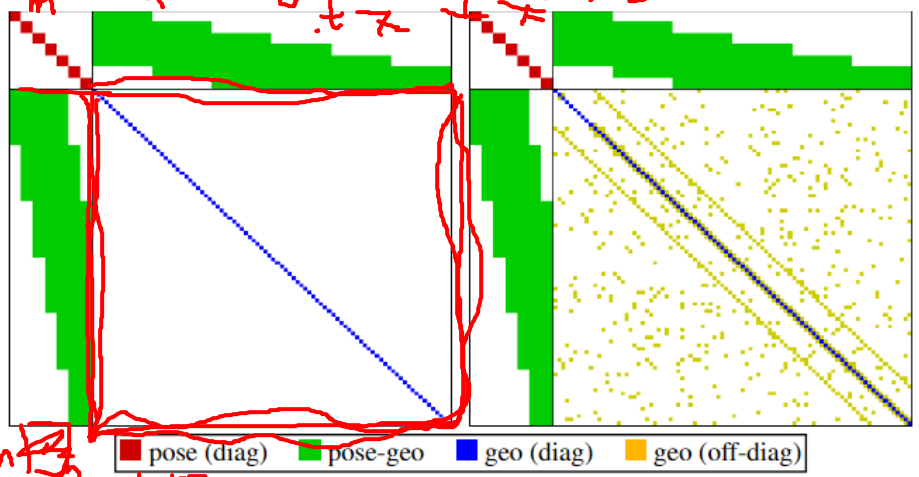
苹果1块一个买x个
梨两块1个买y个

$$2x + 2y = 6$$
$$x + 2y = 5$$

$$\begin{bmatrix} \mathbf{H}_{\alpha\alpha} & \mathbf{H}_{\alpha\beta} \\ \mathbf{H}_{\beta\alpha} & \mathbf{H}_{\beta\beta} \end{bmatrix} \begin{bmatrix} \boldsymbol{x}_{\alpha} \\ \boldsymbol{x}_{\beta} \end{bmatrix} = \begin{bmatrix} \mathbf{b}'_{\alpha} \\ \mathbf{b}'_{\beta} \end{bmatrix}, \tag{16}$$

$$\widehat{\mathbf{H}_{\alpha\alpha}} = \mathbf{H}_{\alpha\alpha} - \mathbf{H}_{\alpha\beta}\mathbf{H}_{\beta\beta}^{-1}\mathbf{H}_{\beta\alpha} \tag{17}$$

$$\widehat{\mathbf{b}'_{\alpha}} = \mathbf{b}'_{\alpha} - \mathbf{H}_{\alpha\beta}\mathbf{H}_{\beta\beta}^{-1}\mathbf{b}'_{\beta}. \tag{18}$$



复杂度分析？好处在哪？

那现在H是怎么来的？

$$\mathbf{J}_k = \Big[\underbrace{\frac{\partial I_j}{\partial \mathbf{p}'}}_{\mathbf{J}_I} \underbrace{\frac{\partial \mathbf{p}'((\boldsymbol{\delta}+\boldsymbol{x}) \boxplus \boldsymbol{\zeta}_0)}{\partial \boldsymbol{\delta}_{\text{geo}}}}_{\mathbf{J}_{\text{geo}}}, \underbrace{\frac{\partial r_k((\boldsymbol{\delta}+\boldsymbol{x}) \boxplus \boldsymbol{\zeta}_0)}{\partial \boldsymbol{\delta}_{\text{photo}}}}_{\mathbf{J}_{\text{photo}}} \Big],$$

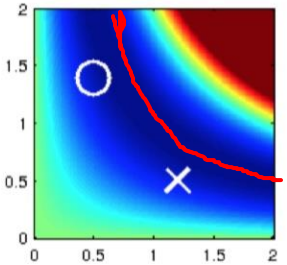
(13)

https://www.cnblogs.com/JingeTU/p/8203606.html https://zhuanlan.zhihu.com/p/77002568

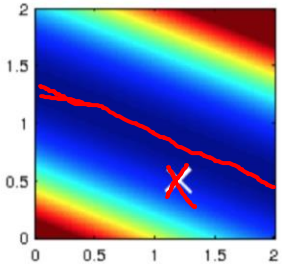
边缘化舒尔补

Windowed, real-time optimization: Consistency.

(for now, let's assume we have initializations, and know which points to use and where they are visible.)

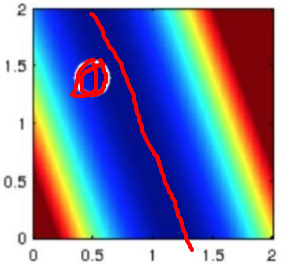


$$E = E_1 + E_2 = (xy-1)^2 + (xy-1)^2$$



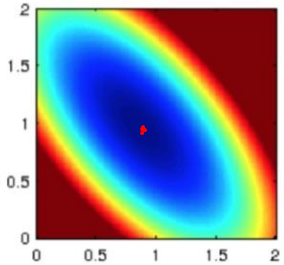
$E_1' = E_1$

Lin. around (0.5, 1.4)



$E_2' = E_2$

Lin. around (1.2, 0.5)



$E' = E_1' + E_2'$

nullspace disappears!

never combine linearizations around different linearization points,
especially in the presence of non-linear nullspaces!
It will render unobservable dimensions observable, and corrupt the system.

帧管理

三个步骤

1.Initial Frame Tracking：

运动模型估计；建立金字塔；在每层进行匹配

论文中提到了：点的数量不需要太多，down scale时深度按照附近的点走

论文中没提到但是你读代码时候可以注意的点：优化过程中的深度在不scale之间是如何传递的...

2.Keyframe Creation：

视角变化大，遮挡，曝光时间变化过大

3.Keyframe Marginalization：

保持最新两帧；在新来的帧中几乎看不到了；帧太多的时候按照和最新帧的距离扔一个

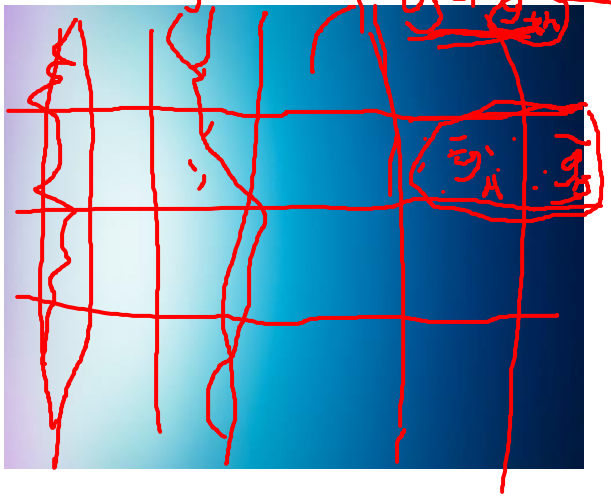


点管理管理

三个步骤

1. **Candidate Point Selection 图上的哪些点?**

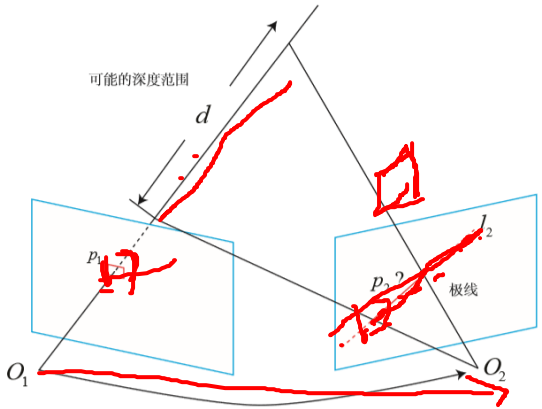
梯度够大；均匀分布　可以用一个阈值，但是如果某些地方特别大怎么办?
如果我们想在白墙上多来点点怎么办?

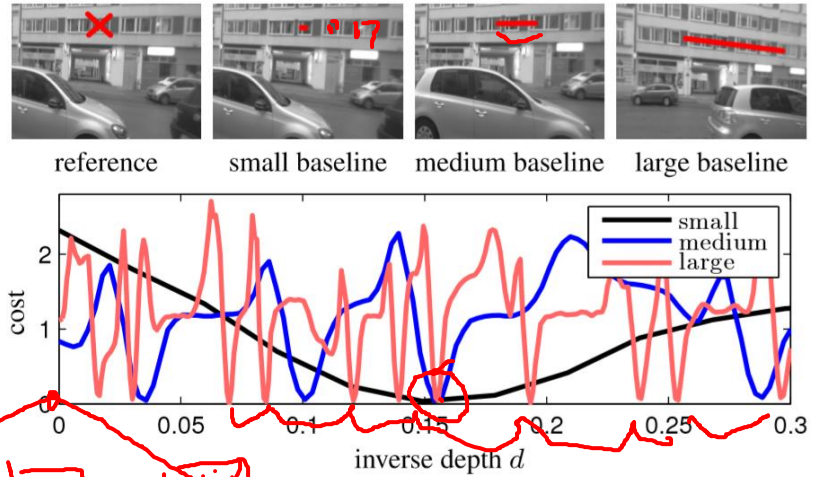




点管理管理

2. Candidate Point Tracking:





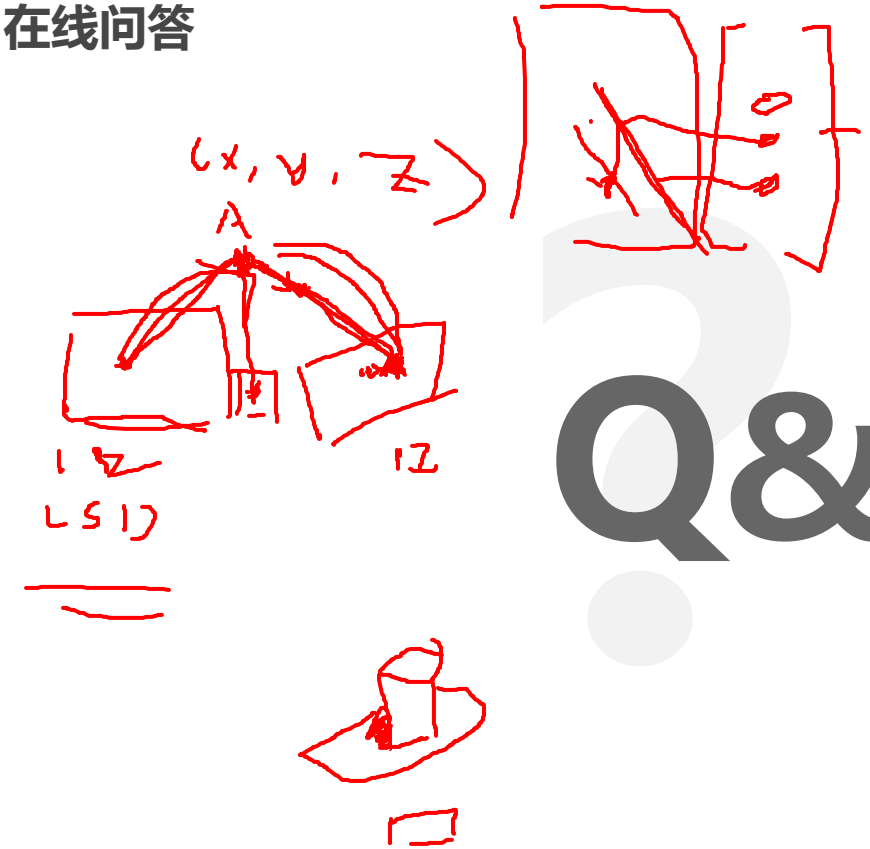
3. Candidate Point Activation:

为了保持点数量，真正优化的点要再选择一下

把以前的点投影的新一帧，选择那些距离投影过来的点远的点

T. Schöps, J. Engel, and D. Cremers. Semi-dense visual odometry for AR on a smartphone. In International Symposium on Mixed and Augmented Reality (ISMAR), 2014.

在线问答



Q&A

感谢各位聆听！

Thanks for Listening

